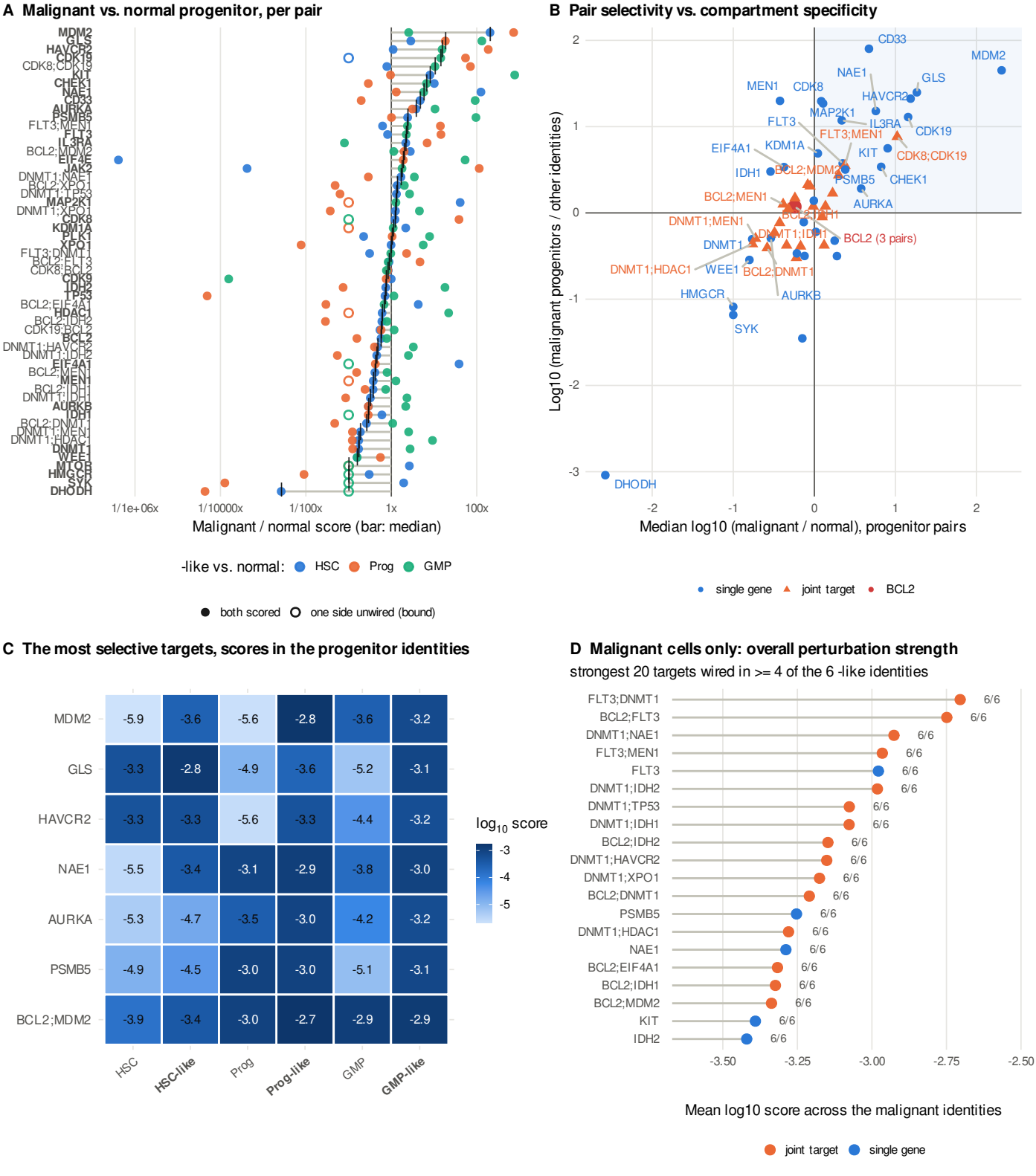
A Malignant vs. normal progenitor, per pair
MDM2
GLS
HAVCR2
CDK19
CDK8;CDK19
KIT
CHEK1
NAE1
CD33
AURKA
PSMB5
FLT3;MEN1
FLT3
IL3RA
BCL2;MDM2
EIF4E
JAK2
DNMT1;NAE1
BCL2;XPO1
DNMT1;TP53
MAP2K1
DNMT1;XPO1
CDK8
KDM1A
PLK1
XPO1
FLT3;DNMT1
BCL2;FLT3
CDK8;BCL2
CDK9
IDH2
TP53
BCL2;EIF4A1
HDAC1
BCL2;IDH2
CDK19;BCL2
BCL2
DNMT1;HAVCR2
DNMT1;IDH2
EIF4A1
BCL2;MEN1
MEN1
BCL2;IDH1
DNMT1;IDH1
AURKB
IDH1
BCL2;DNMT1
DNMT1;MEN1
DNMT1;HDAC1
DNMT1
WEE1
MTOR
HMGCR
SYK
DHODH
1/1e+06x
1/10000x
1/100x
1x
100x
Malignant / normal score (bar: median)
-like vs. normal:
HSC
Prog
GMP
both scored
one side unwired (bound)
B Pair selectivity vs. compartment specificity
Log10 (malignant progenitors / other identities)
Median log10 (malignant / normal), progenitor pairs
CD33
MDM2
NAE1
GLS
MEN1
CDK8
HAVCR2
MAP2K1
FLT3
IL3RA
CDK19
FLT3;MEN1
EIF4A1
KDM1A
KIT
CDK8;CDK19
IDH1
BCL2;MDM2
PSMB5
CHEK1
BCL2;MEN1
AURKA
DNMT1;MEN1
DNMT1;IDH1
BCL2 (3 pairs)
DNMT1
DNMT1;HDAC1
WEE1
BCL2;DNMT1
HMGCR
AURKB
SYK
DHODH
single gene
joint target
BCL2
C The most selective targets, scores in the progenitor identities
MDM2 -5.9 -3.6 -5.6 -2.8 -3.6 -3.2
GLS -3.3 -2.8 -4.9 -3.6 -5.2 -3.1
HAVCR2 -3.3 -3.3 -5.6 -3.3 -4.4 -3.2
NAE1 -5.5 -3.4 -3.1 -2.9 -3.8 -3.0
AURKA -5.3 -4.7 -3.5 -3.0 -4.2 -3.2
PSMB5 -4.9 -4.5 -3.0 -3.0 -5.1 -3.1
BCL2;MDM2 -3.9 -3.4 -3.0 -2.7 -2.9 -2.9
HSC
HSC-like
Prog
Prog-like
GMP
GMP-like
log10 score
-3
-4
-5
D Malignant cells only: overall perturbation strength
strongest 20 targets wired in >= 4 of the 6 -like identities
FLT3;DNMT1 6/6
BCL2;FLT3 6/6
DNMT1;NAE1 6/6
FLT3;MEN1 6/6
FLT3 6/6
DNMT1;IDH2 6/6
DNMT1;TP53 6/6
DNMT1;IDH1 6/6
BCL2;IDH2 6/6
DNMT1;HAVCR2 6/6
DNMT1;XPO1 6/6
BCL2;DNMT1 6/6
PSMB5 6/6
DNMT1;HDAC1 6/6
NAE1 6/6
BCL2;EIF4A1 6/6
BCL2;IDH1 6/6
BCL2;MDM2 6/6
KIT 6/6
IDH2 6/6
-3.50
-3.25
-3.00
-2.75
-2.50
Mean log10 score across the malignant identities
joint target
single gene